### *Change of variable technique*

Class of January 15

In this session we will see the change of variable technique to solve an equation. Let's recall the equation,

$$3^{3x} + 3^x = 3^{2x+1}.$$

We can push the idea of "*factoring*" an expression, in the sense that we can algebraically modify the expression to look for similar terms. With this in mind, let's apply the exponential properties to rewrite the equation,

$$(3^x)^3 + 3^x = (3^x)^2 \cdot 3^1.$$

We can see that in all of the terms appears the factor $3^x$; hence, we can introduce a "*change of variable*" as follows: $y = 3^x$,

$$\begin{aligned} y^3 + y &= y^2 \cdot 3 \\ y^3 + y - 3y^2 &= 0 \end{aligned}$$

Now, instead of solving an exponential equation, we change the problem into finding the roots of a cubic equation.

$$y\big(y^2 - 3y^1 + 1\big) = 0.$$

Hence, one solution is $y = 0$. The other two solutions can be computed using the general formula, considering that $a = 1,\ b = -3,\ c = 1$,

$$\begin{aligned} y &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(1)}}{2(1)(1)} \\ &= \frac{3 \pm \sqrt{5}}{2}. \end{aligned}$$

Now that we found the three roots of the cubic polynomial, $y \in \{0, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\}$, let's finish the equation by reverting the change of variable. For the first solution we have,

$$\begin{aligned} 3^x &= y = 0 \\ \log_3(3^x) &= \log 3(0) \\ x &= \log 3(0). \end{aligned}$$

which does not exits. Therefore, we get only two solutions left,

$$\begin{aligned} 3^x = y &= \frac{3 \pm \sqrt{5}}{2} \\ \log_3(3^x) &= \log_3\left(\frac{3 \pm \sqrt{5}}{2}\right) \\ x &= \log_3\left(\frac{3 \pm \sqrt{5}}{2}\right). \end{aligned}$$

And those are the intersection points of the original equation.

*Class Activities* Solve the following equations

$$\begin{aligned} 3^{2x} + 3^x &= 3^{2x+2} \\ e^{2x} - 3e^x + 2 &= 0 \end{aligned}$$